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Where Minds Pollinate

Class 12
BIOLOGY
Chapter 3
NOTES
NBF book

Follow Our Socials



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Chapter # 3 : Respiratory System of Man

↓
Respiration [Energy Extraction by OXIDATION of food]

External
Respiration
(Physiological)

- Breathing

- INHALATION
EXHALATION

- Energy consuming

Internal
Respiration
(Biochemical)

- Cellular
Respiration

AEROBIC and
ANAEROBIC.

ATP

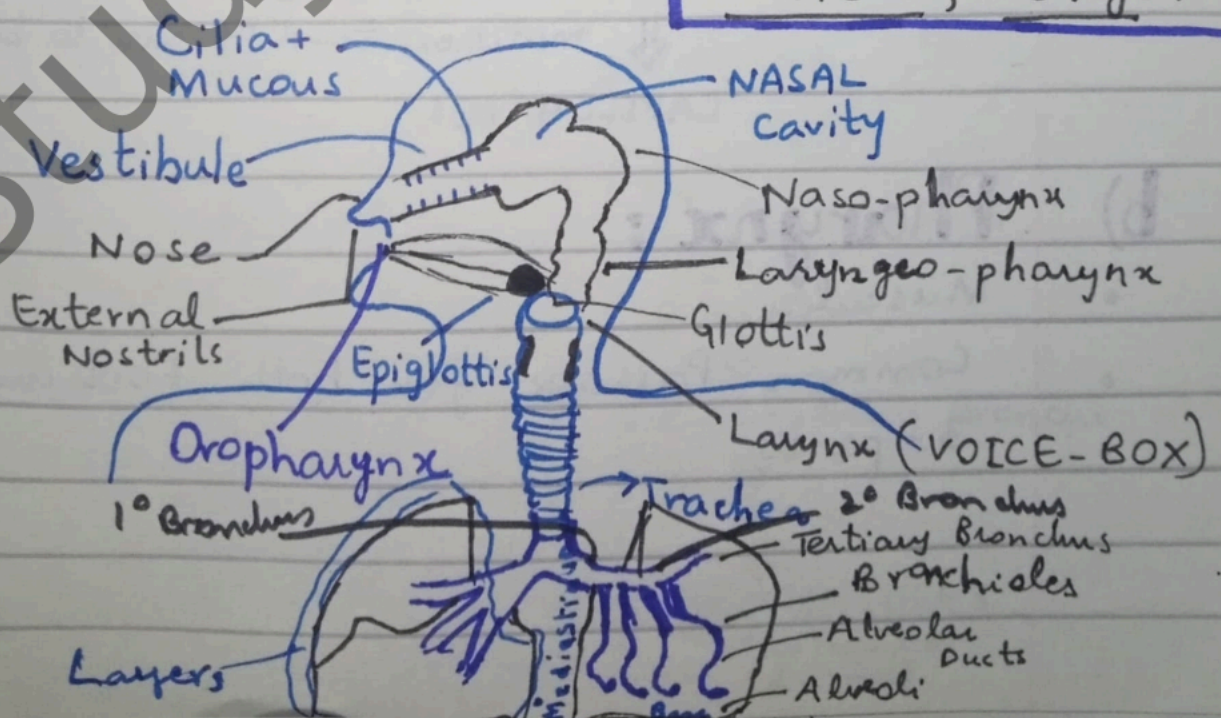
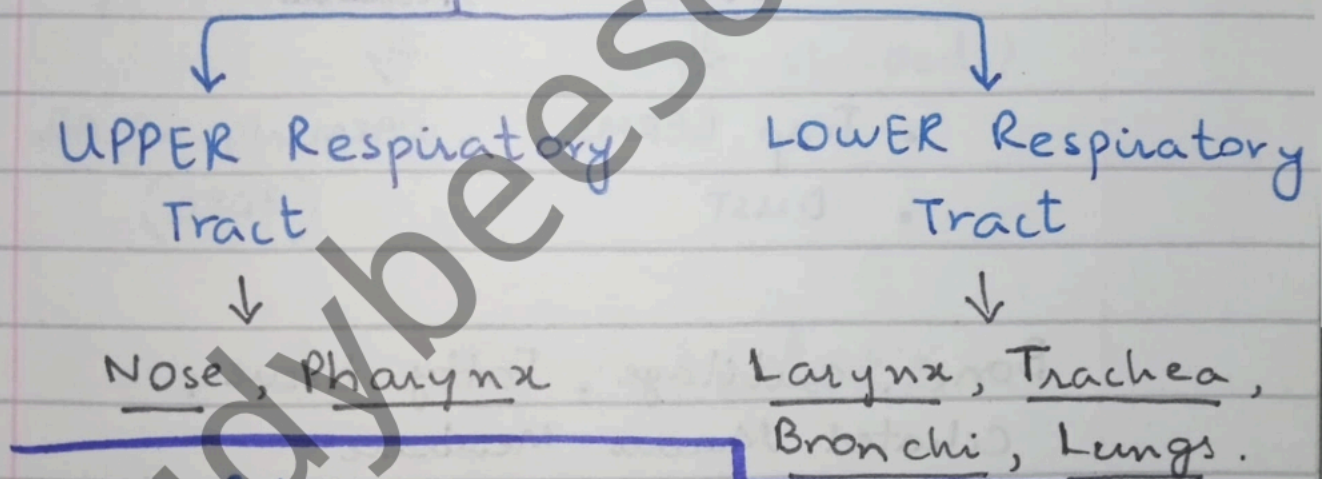
RESPIRATORY SURFACE

↓
Area / Structure where Gaseous
Exchange takes place.

PROPERTIES :

- Should be THIN.
- Permeable
- Moist
- Large surface area.
- Supplied with blood capillaries.
- Should have ventilation mechanism.

Respiratory System



Upper Respiratory Tract :

a) Nose :

- Only EXTERNAL visible portion of Respiratory system.
- Opening → Nostrils
- Space → Nasal - Cavities
- Walls → Vestibule

Cilia



- Trap GERMS
- DUST

Mucous



- WARMING up Air
(30°C)

Bone, cartilage, Fatty-tissue,
Ciliated Mucous Membrane

↓ matches temp of air to be passed to
CAPILLARIES

b) Pharynx :

- Muscular
- Common Passage for both Food and AIR.
- 3 - parts

Lower Respiratory Tract :

a) Larynx : (VOICE - BOX)

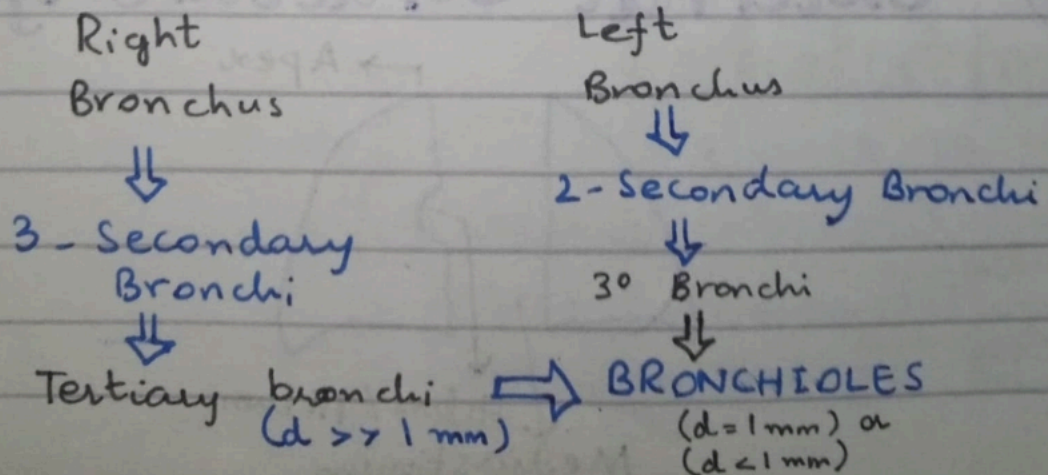
- Smooth Muscle
- Cartilage → EPI GLOTTIS
- Opening → GLOTTIS
- Vocal chord → Vibration
↓
Vocalization

b) Trachea :

- Cartilage (15 - 20 C-shaped)
- wind pipe
- Smooth Muscle

c) Bronchi / Bronchioles :

TRACHEA $\xrightarrow[\text{cavity}]{\text{chest}}$ Divide



d) Alveolar Duct and Alveoli:

Bronchioles

Tube

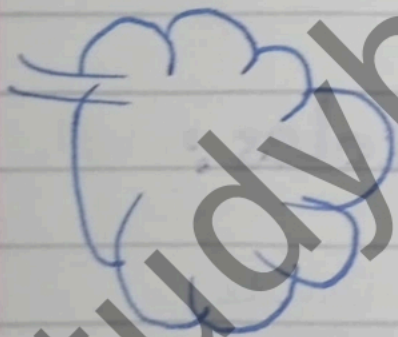
Alveolar Duct

Alveoli

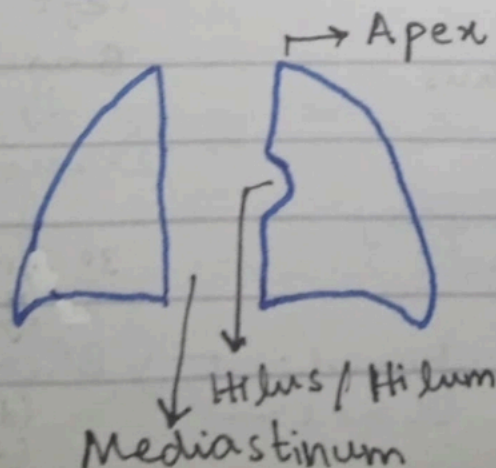
700 million in lungs

0.1 μ m

Richly supplied with
BLOOD CAPILLARIES, COLLAGEN
and ELASTIN.



e) External Structure of Lungs

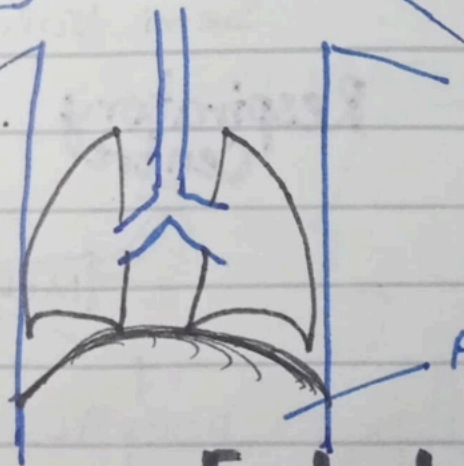


Mechanism of Breathing

THORACIC cavity

Inhalation:

- Intake of O_2
 - External intercostal muscles CONTRACT.
 - Internal intercostal muscles RELAX.
 - Diaphragm contracts.
- ⇒ Becomes flat.



Abdominal cavity

Exhalation:

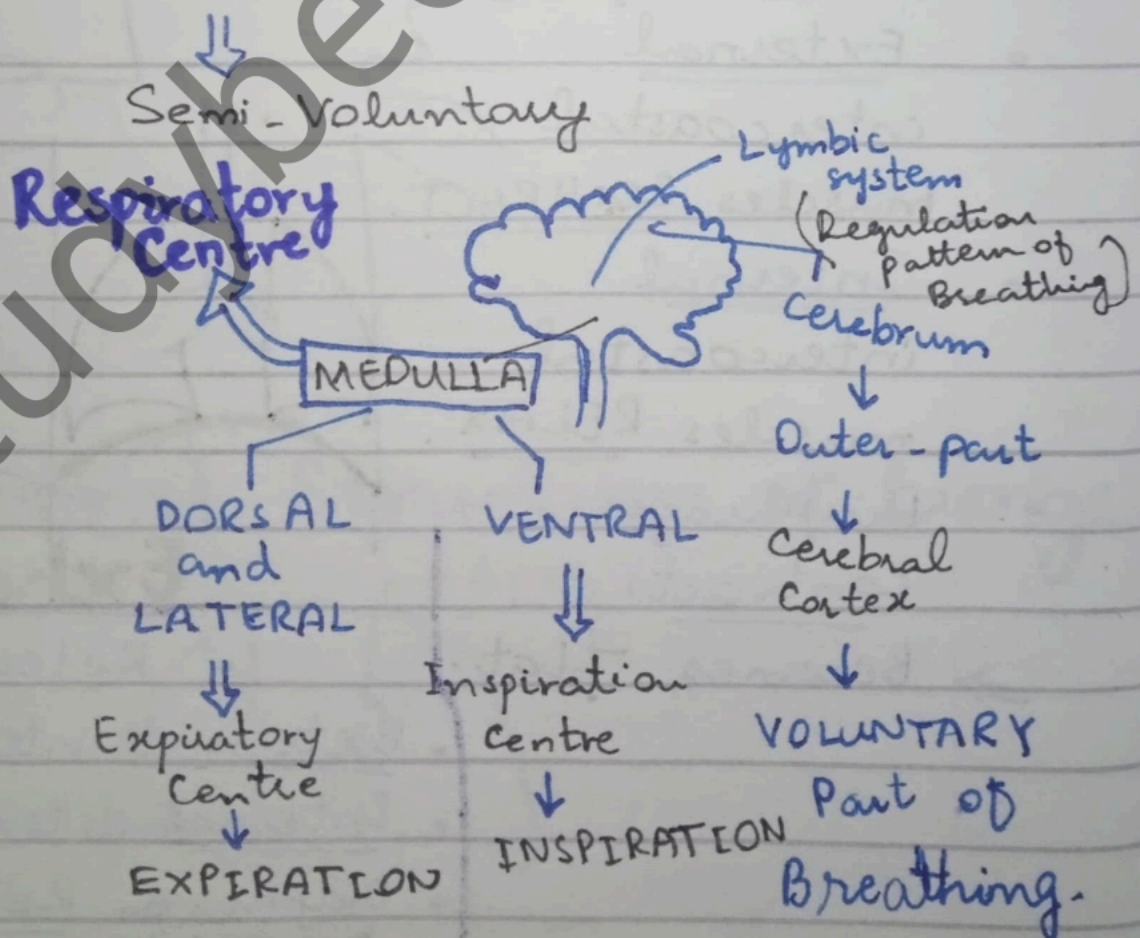
- Release of CO_2 .
- External intercostal muscles RELAX.
- Internal intercostal muscles contract.
- Diaphragm Relaxes. Becomes Dome-like.

- Ribs move FORWARD and UPWARD.
- Volume of chest cavity INCREASES.
- Lungs EXPAND.
- Pressure inside lungs $\rightarrow$ low.

- Ribs move DOWNWARD and BACKWARD.
- Volume of chest cavity DECREASES.
- Lungs COMPRESS.
- Pressure inside lungs $\rightarrow$ high.

Control of Breathing:

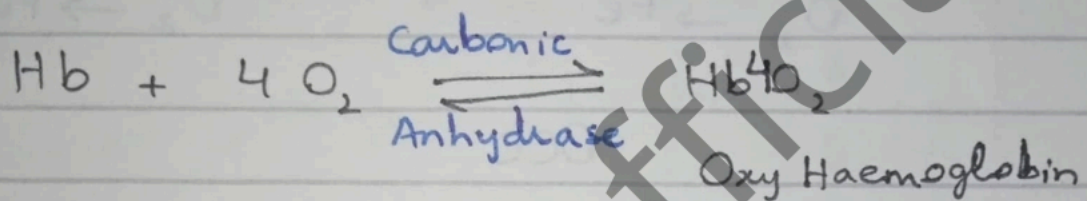
Breath Regulation



Transport of Oxygen

By HAEMOGLOBIN
97%

By PLASMA
3%



15g of Hb = 100ml of Blood

1.34 ml of O_2 = 1g of Hb

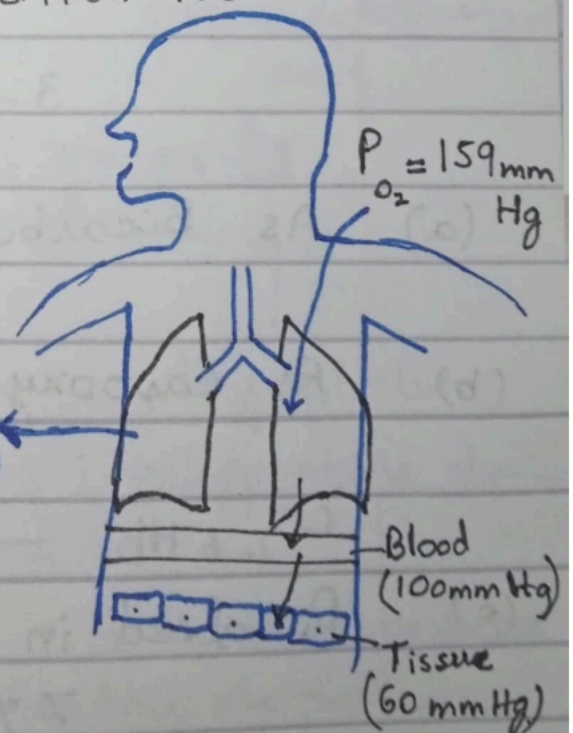
$15 \times 1.34 = 20 \text{ ml of } \text{O}_2 / 100 \text{ ml of blood}$

↓
100% SATURATED

19.6 ml / 100 ml

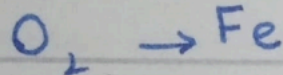
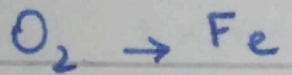
↓
98% SATURATED

$P_{\text{O}_2} = 119 \text{ mm Hg}$
Partial Pressure



Haemoglobin

2- α chains



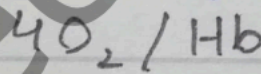
2- β chains



OCC



Oxygen - Carrying Capacity
of Haemoglobin



TRANSPORT OF CO_2

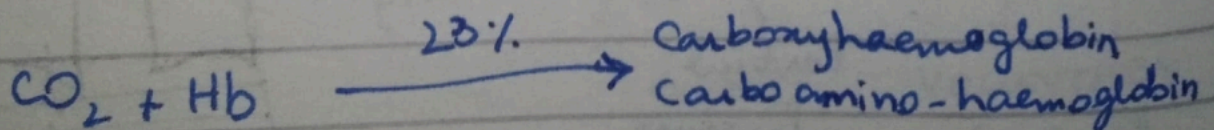


3-Ways

(a) As Bicarbonate ions

70%

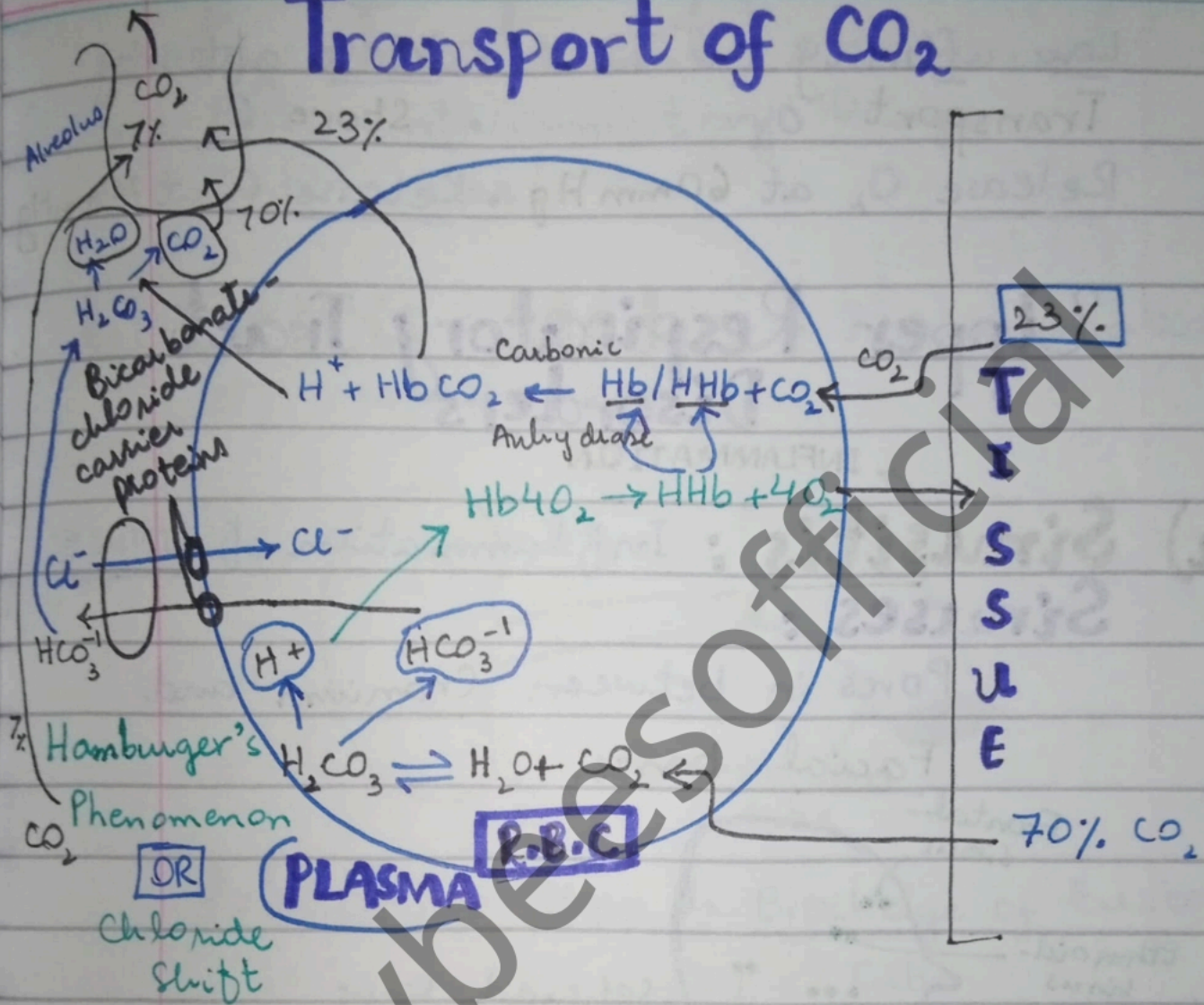
(b) As Carboxyhaemoglobin



(c) Dissolved in PLASMA

7%

Transport of CO₂



HHb = Haemoglobinic-Acid

Respiratory Pigments

Haemoglobin

- 4 - polypeptide chain
- 4 - Fe
- 4 O₂ can combine
- Present in R.B.C.s

Myoglobin

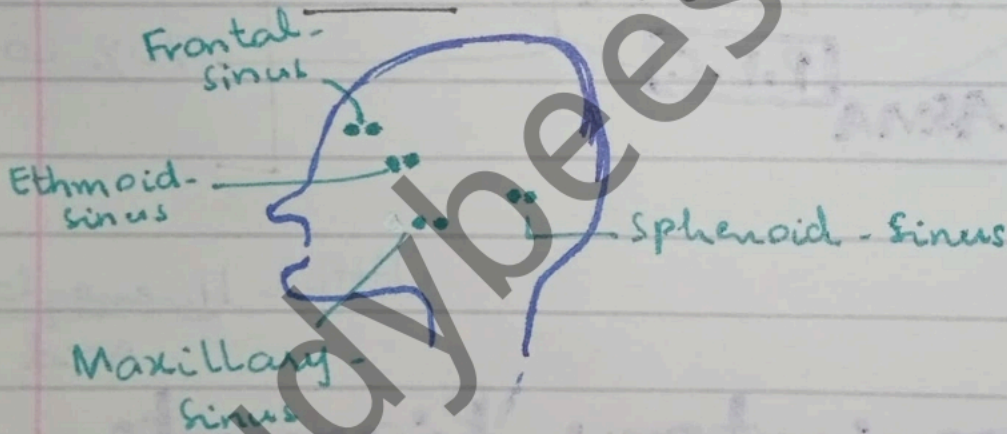
- 1 - polypeptide chain
- 1 - Fe
- 1 O₂ can combine
- Muscle protein

- | | |
|--|---|
| <ul style="list-style-type: none"> • <u>Low</u> affinity • Transport O_2 • <u>Release</u> O_2 at 60mmHg | <ul style="list-style-type: none"> • <u>More</u> affinity • Store O_2 • <u>Release</u> O_2 at 20mmHg |
|--|---|

Upper Respiratory Tract Disorders

INFLAMMATION
 a) **Sinusitis** : Inflammation of SINUSES.
Sinuses :

Pores in between Cranium and Facial bones.



Etiology / Reason :

Pollution, Smoke, Dust, Over-crowding, Dental Infection.

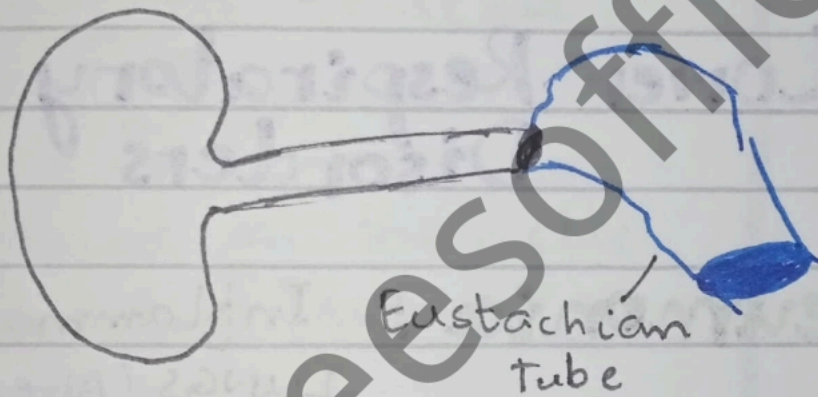
Sign / Symptoms :

- | | |
|---|--|
| <ul style="list-style-type: none"> • Fever • Headache • Nausea • Loss of sense of <u>SMELL</u>. | <ul style="list-style-type: none"> • Raspy Voice • Nasal Obstructions • Nasal Discharge |
|---|--|

R_x (Treatment):

- Anti-inflammatory Drugs.
- Antibiotics

b) Otitis Media : Inflammation of MIDDLE EAR.



Mechanism : Blockage of Eustachian Tube

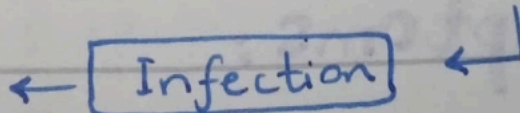


Fluid NOT drain
Accumulation behind Eardrum



Bacteria GROW in fluid

PAIN



Symptoms :

- Pain
- Tugging of Ear
- Difficulty in sleeping and hearing.

R_x :

ANTIBIOTICS in the form of ear drops or tablets.

Etiology:

- Bacteria (Main cause).
- Viral + Fungal infection.

Lower Respiratory Tract Disorders

a) **Pneumonia:** Inflammation of LUNGS (Alveoli)

Mechanism:

Bacteria → Nose → Lungs

Dusky / Purplish Skin

Bacteria stay in alveolus

↑ attack

Poor oxygenation

Pus in alveolus

W.B.Cs

Symptoms:

- Fever
- Chill
- Effect on MENTAL HEALTH
- Nausea
- Vomiting
- Phlegm (Sputum) (بلغم)

Etiology: → { Mycobacterium
Streptococcus Pneumoniae Tuberculosis

Rx:

Antibiotics

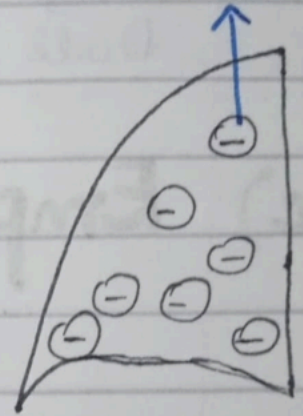
b) Pulmonary Tuberculosis: (T.B)

- Contagious Disease
- Alveoli Damage



Replaced by inelastic
connective tissue

TUBERCLE



Mechanism:

Bacilli attacks on LUNGS.



Covered by Lung's Epithelium



Bacteria replicates in covering



COVER BURST



Bacteria ATTACK on multiple
surface of lungs



TUBERCLE

Symptoms:

- Intermittent fever.
- Light sweat.
- ANOREXIA
- Depression
- Chill
- weakness
- Cough
- Dull ache in chest

R_x:

9 months

DOTS



Daily Observed
Treatment Short-
Course.

c) Emphysema: 'breakdown of alveoli'



No proper oxygenation
Dusky / Purplish

Etiology:

- Smoking
- Cigar
- Vaping
- Pollution
- Dust

Symptoms:

- Shortness of Breath
- Cough
- Difficulty in climbing
up small height/
up-stairs.

d) Chronic Obstructive Pulmonary Disease (COPD):

Collection of Different Lung Diseases

Emphysema

Damage to
air-sacs

Chronic
Bronchitis

Sign / Symptoms:

- Shortness of breath
- Cough with phlegm
- Infection
- wheezing

Rx:

- INHALERS.